

View Review

Paper ID

156

Paper Title

An Integrated Framework for Load Balancing in Cloud Computing: Adaptive Heuristic, Machine Learning, and Energy-Aware Approaches for Optimize Resource

Track Name

Hardware/Software Solutions for AI & Big Data (SAB)

REVIEW QUESTIONS**1. Overall evaluation:**

accept

2. Reviewer's confidence

High

3. Relevance of the contribution for IEEE-iSES 2025

highly relevant

4. Contribution to the computing education research field

a clear contribution to the field

5. Likelihood of generating discussion at the conference that will benefit the field

extremely likely

6. Best Paper Recommendation?

Yes

7. Paper for poster présentation recommandation?

No

8. Paper comments:

- ☐ Integrated Framework: The paper's core strength is its integrated framework, which combines an Adaptive Heuristic Load Balancing Algorithm (AH-LBA), a Machine Learning-based Load Prediction Algorithm (ML-LPA), and an Energy-Aware Load Optimization Algorithm (EA-LOA).
- ☐ Superior Performance: The proposed AH-LBA significantly outperformed existing algorithms, achieving a reduced load variance of 12.67 and a makespan of 298 ms under high workloads.
- ☐ High Prediction Accuracy: The ML-LPA, which uses LSTM models, demonstrated superior workload prediction accuracy of up to 94.3%.
- ☐ Energy Efficiency: The EA-LOA achieved the lowest energy consumption at just 7.8 kWh under high workloads, highlighting its effectiveness in minimizing power usage.